Recurrence Relations

- Very frequent in algorithm analysis
  - Recursive algorithms

- Recurrence relation
  - Recurrence equation
  - Recurrence
  - Recursive relation
  - Difference equation

- Solving a recurrence relation
  - Converting to closed-form

Recurrence Relations

- Type 1
  - $x(n) = c_1 * x(n-1) + c_2 * x(n-2) + \cdots + c_k * x(n-k) + f(n)$
    $x(0), \ldots, x(k-1)$ given

    k^{th}-order recurrence relation

- Type 2
  - $x(n) = c * x\left(\frac{n}{b}\right) + f(n)$
    $x(1)$ given

- May be different initials
- Requirement:
  Sufficient number of initials

Recurrence Relations

- Type 1
  - $x(n) = c_1 * x(n-1) + c_2 * x(n-2) + \cdots + c_k * x(n-k) + f(n)$
    $x(0), \ldots, x(k-1)$ given

    ⟹ k^{th}-order recurrence relation

- Type 2
  - $x(n) = c * x\left(\frac{n}{b}\right) + f(n)$
    $x(1)$ given

- Linear
  $a_1 x_1 + \cdots + a_n x_n + b = 0$
  $x(n) = c_1 x^2(n-1) + \cdots + f(n)$ ✗
- Homogeneous – Inhomogeneous
  $f(n) = 0$ – $f(n) \neq 0$
- $c_1, \ldots, c_k \rightarrow$ constant

Recurrence Relations

- Type 1
  - $x(n) = c_1 * x(n-1) + c_2 * x(n-2) + \cdots + c_k * x(n-k) + f(n)$

    $x(0), \ldots, x(k-1)$ given

    ⟹ k^{th}-order recurrence relation

- Type 2
  - $x(n) = c * x\left(\frac{n}{b}\right) + f(n)$

    $x(1)$ given

complexity of the algorithm for input size *n*

calls itself recursively with input sizes *n*-1, *n*-2, ..., *n*-*k*, *n*/*b*
Complexity for that input size

number of times of the recursive call

work in addition to the recursive calls

Recurrence Relations

- Forward substitution
  - Start with initial terms
  - Generate first few terms



  - Obtain a general pattern in closed-form
  - Check validity
    - by direct substitution
    - by mathematical induction

Recurrence Relations

- Forward substitution
  - Example

$$x(n) = 2x(n-1) + 1 \ , \quad x(1) = 1$$

$$x(1) = 1$$
$$x(2) = 2x(1) + 1 = 3$$
$$x(3) = 2x(2) + 1 = 7$$
$$x(4) = 2x(3) + 1 = 15$$

n : 1 2 3 4 …
sequence : 1, 3, 7, 15, …

$$\Rightarrow x(n) = 2^n - 1 \ , n \geq 1$$

Check validity of the "guessed solution"

$$x(n) = 2x(n-1) + 1 \Rightarrow 2^n - 1 = 2 * (2^{n-1} - 1) + 1 \Rightarrow 2^n - 1 = 2^n - 1$$



Recurrence Relations

- Forward substitution
  - Start with initial terms
  - Generate first few terms



  - Obtain a general pattern in closed-form
  - Check validity
    - by direct substitution
    - by mathematical induction

Works in a very limited number of cases
Difficult to recognize a pattern

Recurrence Relations

- Backward substitution
  - Express $x(...)$ terms on RHS with lower-order terms
  - Substitute into original relation

  } Repeat until initials

  ↓
  - Obtain a pattern & Express in terms of summation

  ↓
  - Convert into closed-form
    - by summation formulae
    - by integration technique
    - …

Works surprisingly well for a wide variety of simple recurrences

Recurrence Relations

- Backward substitution
  - Express $x(\ldots)$ terms on RHS with lower-order terms
  - Substitute into original relation

  } Repeat until initials

  ↓
  - Obtain a pattern & Express in terms of

  ↓
  - Convert into closed-form
    - by summation formulae
    - by integration technique
    - …

$$
\begin{aligned}
T(n) &= 2 * T(n-1) + 1 \\
&= 2 * [2 * T(n-2) + 1] + 1 = 2^2 * T(n-2) + 2 + 1 \\
&= 2^2 * [2 * T(n-3) + 1] + 2 + 1 = 2^3 * T(n-3) + 2^2 + 2 + 1 \\
&\ldots \\
&= 2^{n-1} * T(1) + 2^{n-2} + 2^{n-3} + \cdots + 2 + 1 \\
&= 2^{n-1} + 2^{n-2} + 2^{n-3} + \cdots + 2 + 1 \\
&= \textstyle\sum_{i=0}^{n-1} 2^i \\
\\
\Rightarrow T(n) &= 2^n - 1
\end{aligned}
$$

Works surprisingly well for a wide variety of simple recurrences

Recurrence Relations

- Change of variable
  - Substitution does not always take us to a solution
  - Example

$$x(n+1) = b_{n+1}x(n) + c_{n+1} \quad (n \geq 0)\,, \quad x(0) \text{ given}$$

Backward substitution

$$x(n+1) = b_{n+1}[b_n x(n-1) + c_n] + c_{n+1}$$
$$\dots$$
$$= (b_{n+1} \dots b_1)x(0) + (b_{n+1} \dots b_2)c_1 + (b_{n+1} \dots b_3)c_2 + \cdots + c_{n+1}$$



Difficult to convert to closed-form

Recurrence Relations

- Change of variable
  - Example

$$x(n+1) = b_{n+1}x(n) + c_{n+1} \quad (n \geq 0)\,, \quad x(0) \text{ given}$$

problem: coefficient modifying the recursive term on RHS

$x(n+1) = b_{n+1}x(n)$ ✗

$y(n+1) = y(n)$ ✓

  - Step 1: Make a change of variable that leads to a new recurrence

$$x(n) = b_1 b_2 \ldots b_n y(n) \quad (n \geq 1)\,, \quad y(1) = (x(1)/b_1) \text{ given}$$

$$\Rightarrow b_1 b_2 \ldots b_{n+1} y(n+1) = b_{n+1} b_1 b_2 \ldots b_n y(n) + c_{n+1}$$

$$\Rightarrow y(n+1) = y(n) + d_{n+1} \text{ , where } d_{n+1} = \frac{c_{n+1}}{b_1 b_2 \ldots b_{n+1}}$$

Recurrence Relations

- Change of variable
  - Example

$$x(n+1) = b_{n+1}x(n) + c_{n+1} \quad (n \geq 0), \quad x(0) \text{ given}$$

problem: coefficient modifying the recursive term on RHS

$x(n+1) = b_{n+1}x(n)$ ✗

$y(n+1) = y(n)$ ✓

  - Step 2: Solve the new recurrence

$$y(n+1) = y(n) + d_{n+1}$$

$\ldots$ $\Longrightarrow$ backward substitution

$$y(n) = y(1) + \sum_{i=2}^{n+1} d_i$$

Recurrence Relations

- Change of variable
  - Example

$$x(n+1) = b_{n+1}x(n) + c_{n+1} \quad (n \geq 0), \quad x(0) \text{ given}$$

problem: coefficient modifying the recursive term on RHS

$x(n+1) = b_{n+1}x(n)$ ✗

$y(n+1) = y(n)$ ✓

  - Step 3: Return to the original variable

$$x(n) = b_1 b_2 \ldots b_n y(n)$$

$$\Rightarrow x(n) = b_1 b_2 \ldots b_n \left[x(1)/b_1 + \sum_{i=2}^{n+1} d_i\right]$$

Convert to closed-form

Recurrence Relations

- Change of variable
  - Example

$$x(n+1) = 3x(n) + n \quad (n \geq 0), \quad x(0) = 0$$

  - Step 1: Make a change of variable

$$x(n) = 3^n y(n)$$

$$\Rightarrow y(n+1) = y(n) + \frac{n}{3^{n+1}} \quad (n \geq 0), \quad y(0) = 0$$

Backward substitution works? ✔

$$x(n+1) = n\sum_{i=0}^{n} 3^i - \sum_{i=1}^{n} i3^i$$

  - Step 2: Solve

$$y(n) = \sum_{i=1}^{n-1} \frac{i}{3^{i+1}}$$

  - Step 3: Return to the original variable

$$x(n) = 3^n y(n) = 3^n \sum_{i=1}^{n-1} \frac{i}{3^{i+1}} \quad \Longrightarrow \quad \text{Closed-form?}$$

Recurrence Relations

- Characteristic equation
  - 2nd-order recurrence relations

$$x(n+1) = a * x(n) + b * x(n-1) \quad (n \geq 1), \quad x(0) \,\&\, x(1) \text{ given}$$

  - Substitution 

Recurrence Relations

- Characteristic equation
  - 2nd-order recurrence relations

    $x(n+1) = a * x(n) + b * x(n-1) \quad (n \geq 1), \quad x(0)$ & $x(1)$ given
  - Characteristic equation: $\beta^2 = a\beta + b$
    - Known: $x(n) = \beta^n$ solves this recurrence

      Test it (two pages later)

      Substitute $f(n) = \beta^n$, with $\beta = \frac{1+\sqrt{5}}{2}$

      $\Rightarrow 2.61 = 2.61$
    - Substitute into the recurrence

      $\beta^{n+1} = a * \beta^n + b * \beta^{n-1}$

      $\Rightarrow \beta^2 = a * \beta + b$
    - Now, find suitable values of β

Recurrence Relations

- Characteristic equation

  - 2nd-order recurrence relations

    $x(n+1) = a * x(n) + b * x(n-1) \quad (n \geq 1)\,, \quad x(0)$ & $x(1)$ given

  - Characteristic equation: $\beta^2 = a\beta + b$

    ↓

  - Find roots (β_+ & β_-)

    → 1. β_+ & β_- real and distinct — $x(n) = c_1{\beta_+}^n + c_2{\beta_-}^n$

    → 2. $\beta_+ = \beta_-$ (say, β) and real — $x(n) = c_1\beta^n + c_2 n\beta^n$

    → 3. β_+ & β_- complex — irrelevant

Recurrence Relations

- Characteristic equation
  - Example: Fibonacci numbers

$$f(n+1) = f(n) + f(n-1) \quad (n \geq 1), \quad f(0) = 0 \;\&\; f(1) = 1$$

Characteristic equation: $\beta^2 = \beta + 1$

Roots: $\beta_+ = \frac{1+\sqrt{5}}{2}$ & $\beta_- = \frac{1-\sqrt{5}}{2}$

$$\Rightarrow f(n) = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$\left.\begin{aligned} f(0) &= 0 = c_1 + c_2 \\ f(1) &= 1 = c_1 \left(\frac{1+\sqrt{5}}{2}\right) + c_2 \left(\frac{1-\sqrt{5}}{2}\right) \end{aligned}\right\} \quad c_1 = \frac{1}{\sqrt{5}}, \; c_2 = -\frac{1}{\sqrt{5}}$$

$$\Rightarrow f(n) = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

Recurrence Relations

- Characteristic equation
  - Example:

$$x(n+1) = 2x(n) - x(n-1) \quad (n \geq 1), \quad x(0) = 1 \;\&\; x(1) = 5$$

Characteristic equation: $\beta^2 = 2\beta - 1$

Roots: $\beta_+ = \beta_- = 1 \quad (\beta)$

$$\Rightarrow x(n) = c_1 + c_2 n$$

$$\left.\begin{aligned} x(0) &= 1 = c_1 \\ x(1) &= 5 = c_1 + c_2 \end{aligned}\right\} \quad c_1 = 1 \;\; , \;\; c_2 = 4$$

$$\Rightarrow x(n) = 4n + 1$$

Recurrence Relations

- How about
  - $A(n) = A(n-1) + 3\log n$ ⟹ $\theta(n \log n)$
  - $B(n) = 3B\left(\frac{n}{2}\right) + 5$ ⟹ $\theta(n^{1.58})$
  - $W(n) = 2W\left(\frac{n}{4}\right) + 5n^2 + n$ ⟹ $\theta(n^2)$

Recurrence Relations

- Type 1
  - $x(n) = ax(n-1) + f(n)\ ,\quad x(1) = b$



"decrease by one" algorithms

Exploiting a relationship between a given instance of size *n* and a smaller instance of size *n*-1

Example: Factorial calculation, Insertion sort

Recurrence Relations

- Type 2
  - $x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (b > 1)$

Reducing a given instance of size *n* to one/more smaller instance(s) of size n/*b*



$a = 1 \rightarrow$ "decrease by a constant factor" algorithms

Example: Binary search



$a > 1 \rightarrow$ "divide and conquer" algorithms

Example: Merge sort

Recurrence Relations

- Type 1
  - $x(n) = ax(n-1) + f(n)\ ,\quad x(1) = b$

    $\Rightarrow x(n) = a^{n-1}b + \sum_{i=2}^{n} a^{n-i} f(i)$ (backward substitution)

    - $a = 1\ ,\ f(n) = 1$

      $\Rightarrow x(n) = b + (n-1) \in \theta(n)$

    - $a = 1\ ,\ f(n) = \log n$

      $\Rightarrow x(n) = b + \sum_{i=2}^{n} \log i \in \theta(n \log n)$

Recurrence Relations

- Type 2
  - $x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (b > 1)$

- Equation in the form $x(n) = a * x\left(\frac{n}{b}\right) + \cdots$

  n must be b^k

  θ-invariant under scaling $\Rightarrow n \in \mathbb{N}$

- Equation in the form $x(n) = a * x\left(\left\lceil\frac{n}{b}\right\rceil\right) + \cdots$

  $n \in \mathbb{N}$

Recurrence Relations

- Type 2

  - $x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (b > 1 \, , n = b^k)$

$\Rightarrow x(n) = a^k c + \sum_{i=0}^{k-1} a^i f\left(\frac{n}{b^i}\right)$ (backward substitution)

$f(n) = d$

$\Rightarrow x(n) = a^k c + d \sum_{i=0}^{k-1} a^i$

$\sum_{i=0}^{n-1} x^i = \frac{x^n - 1}{x - 1} \, , x \neq 1 \, , n \in \mathbb{N}^+$

$a^k = a^{\log_b n} = \left(b^{\log_b a}\right)^{\log_b n} = \left(b^{\log_b n}\right)^{\log_b a} = n^{\log_b a}$

$$x(n) = \begin{cases} n^{\log_b a} c + d\left(\frac{n^{\log_b a} - 1}{a - 1}\right) = \left(c + \frac{d}{a - 1}\right) n^{\log_b a} - \frac{d}{a - 1} \in \theta\left(n^{\log_b a}\right) & , a \neq 1 \\ c + d\,k = c + d \log_b n \in \theta(\log n) & , a = 1 \end{cases}$$

Recurrence Relations

- Type 2

  - $x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (b > 1 \, , n = b^k)$

$$\Rightarrow x(n) = a^k c + \sum_{i=0}^{k-1} a^i f\left(\frac{n}{b^i}\right)$$ (backward substitution)

$$\hookrightarrow \quad f(n) = dn$$

$$\Rightarrow x(n) = a^k c + \sum_{i=0}^{k-1} a^i \frac{dn}{b^i} = a^k c + dn \sum_{i=0}^{k-1} \left(\frac{a}{b}\right)^i$$

$$\begin{cases} n^{\log_b a} c + dn \dfrac{\left(\frac{a}{b}\right)^k - 1}{\left(\frac{a}{b}\right) - 1} = n^{\log_b a} c + d \dfrac{n^{\log_b a} - n}{\left(\frac{a}{b}\right) - 1} = \begin{cases} n^{\log_b a}\left(c + \dfrac{d}{a/b - 1}\right) - \dfrac{dn}{a/b - 1} \in \theta\left(n^{\log_b a}\right) & , a > b \\ n^{\log_b a}\left(c + \dfrac{d}{a/b - 1}\right) + \dfrac{dn}{1 - a/b} \in \theta(n) & , a < b \end{cases} \\ n^{\log_b a} c + dn \displaystyle\sum_{i=0}^{k-1} 1 = nc + dn \log_b n \in \theta(n \log n) & , a = b \end{cases}$$

Recurrence Relations

- Type 2
  - $x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (b > 1 \, , n = b^k)$

$$x(n) = \begin{cases} n^{\log_b a} c + d\left(\frac{n^{\log_b a} - 1}{a - 1}\right) = \left(c + \frac{d}{a-1}\right) n^{\log_b a} - \frac{d}{a-1} \in \theta\left(n^{\log_b a}\right) & , a \neq 1 \\ c + d\,k = c + d \log_b n \in \theta(\log n) & , a = 1 \end{cases}$$

Valid for $n = b^k$

$$x(n) = \begin{cases} n^{\log_b a} c + dn \frac{\left(\frac{a}{b}\right)^k - 1}{\left(\frac{a}{b}\right) - 1} = n^{\log_b a} c + d \frac{n^{\log_b a} - n}{\left(\frac{a}{b}\right) - 1} = \begin{cases} n^{\log_b a}\left(c + \frac{d}{a/b - 1}\right) - \frac{dn}{a/b - 1} \in \theta\left(n^{\log_b a}\right) & , a > b \\ n^{\log_b a}\left(c + \frac{d}{a/b - 1}\right) + \frac{dn}{1 - a/b} \in \theta(n) & , a < b \end{cases} \\ n^{\log_b a} c + dn \sum_{i=0}^{k-1} 1 = nc + dn \log_b n \in \theta(n \log n) & , a = b \end{cases}$$

- θ-invariant under scaling
- eventually nondecreasing

$\Longrightarrow$ Valid for $n \in \mathbb{N}$

Recurrence Relations

- Master Theorem

  - Let $x(n)$ be an eventually nondecreasing function that satisfies the recurrence relation

$$x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (n = b^k \, , b > 1 \, , a \geq 1 \, , k \geq 1)$$

If $f(n) \in \theta(n^p)$, where $p \geq 0$, then

$$x(n) \in \begin{cases} \theta(n^p) & , \quad a < b^p \\ \theta(n^p \log n) & , \quad a = b^p \\ \theta\left(n^{\log_b a}\right) & , \quad a > b^p \end{cases} \quad , for \; n \in \mathbb{N}$$

Recurrence Relations

- Master Theorem

  - Let $x(n)$ be an eventually nondecreasing function that satisfies the recurrence relation

$$x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (n = b^k \, , b > 1 \, , a \geq 1 \, , k \geq 1)$$

If $f(n) \in \theta(n^p)$, where $p \geq 0$, then

$$x(n) \in \begin{cases} \theta(n^p) & , \quad a < b^p \\ \theta(n^p \log n) & , \quad a = b^p \\ \theta\left(n^{\log_b a}\right) & , \quad a > b^p \end{cases} \quad , for \; n \in \mathbb{N}$$

Check our results: $f(n) = d \Rightarrow p = 0$

$$x(n) = \begin{cases} n^{\log_b a} c + d\left(\frac{n^{\log_b a} - 1}{a - 1}\right) = \left(c + \frac{d}{a-1}\right) n^{\log_b a} - \frac{d}{a-1} \in \theta\left(n^{\log_b a}\right) & , a \neq 1 \\ c + d\,k = c + d \log_b n \in \theta(\log n) & , a = 1 \end{cases}$$

Recurrence Relations

- Master Theorem

  - Let $x(n)$ be an eventually nondecreasing function that satisfies the recurrence relation

$$x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (n = b^k \, , b > 1 \, , a \geq 1 \, , k \geq 1)$$

If $f(n) \in \theta(n^p)$, where $p \geq 0$, then

$$x(n) \in \begin{cases} \theta(n^p) & , \quad a < b^p \\ \theta(n^p \log n) & , \quad a = b^p \\ \theta\left(n^{\log_b a}\right) & , \quad a > b^p \end{cases} \quad , for \; n \in \mathbb{N}$$

Check our results: $f(n) = dn \Rightarrow p = 1$

$$x(n) = \begin{cases} n^{\log_b a} c + dn \frac{\left(\frac{a}{b}\right)^k - 1}{\left(\frac{a}{b}\right) - 1} = n^{\log_b a} c + d \frac{n^{\log_b a} - n}{\left(\frac{a}{b}\right) - 1} = \begin{cases} n^{\log_b a}\left(c + \frac{d}{a/b - 1}\right) - \frac{dn}{a/b - 1} \in \theta\left(n^{\log_b a}\right) & , a > b \\ n^{\log_b a}\left(c + \frac{d}{a/b - 1}\right) + \frac{dn}{1 - a/b} \in \theta(n) & , a < b \end{cases} \\ n^{\log_b a} c + dn \sum_{i=0}^{k-1} 1 = nc + dn \log_b n \in \theta(n \log n) & , a = b \end{cases}$$

Recurrence Relations

- Master Theorem

  - Let $x(n)$ be an eventually nondecreasing function that satisfies the recurrence relation

$$x(n) = a * x\left(\frac{n}{b}\right) + f(n) \quad , \quad x(1) = c \quad (n = b^k \, , b > 1 \, , a \geq 1 \, , k \geq 1)$$

If $f(n) \in \theta(n^p)$, where $p \geq 0$, then

$$x(n) \in \begin{cases} \theta(n^p) & , \quad a < b^p \\ \theta(n^p \log n) & , \quad a = b^p \\ \theta\left(n^{\log_b a}\right) & , \quad a > b^p \end{cases} \quad , for \; n \in \mathbb{N}$$

Valid also for O and Ω